

Measurement of surface finish

Introduction

- ▶ Precise measurement is dependent on surface texture
- ▶ Whatever may be the manufacturing process used, it is not possible to produce perfectly smooth surface.
- ▶ The irregularities on the surface are in the form of succession of hills and valleys varying in height and spacing.

Factors affecting surface roughness

- ▶ Vibrations
- ▶ Material of work piece
- ▶ Type of machining
- ▶ Rigidity of the system
- ▶ Type, form, material and sharpness of cutting tools
- ▶ Cutting conditions like feed, speed and depth of cut
- ▶ Type of coolant used

Reasons for controlling surface texture are

- ▶ To improve the service life of components
- ▶ To improve fatigue resistance
- ▶ To reduce initial wear of parts
- ▶ To have close dimensional tolerance on the parts
- ▶ To reduce frictional wear

Classification of geometrical irregularities

- ▶ First order irregularity

Irregularities arising out of inaccuracies in the machine tool itself.

- ▶ Second order irregularity

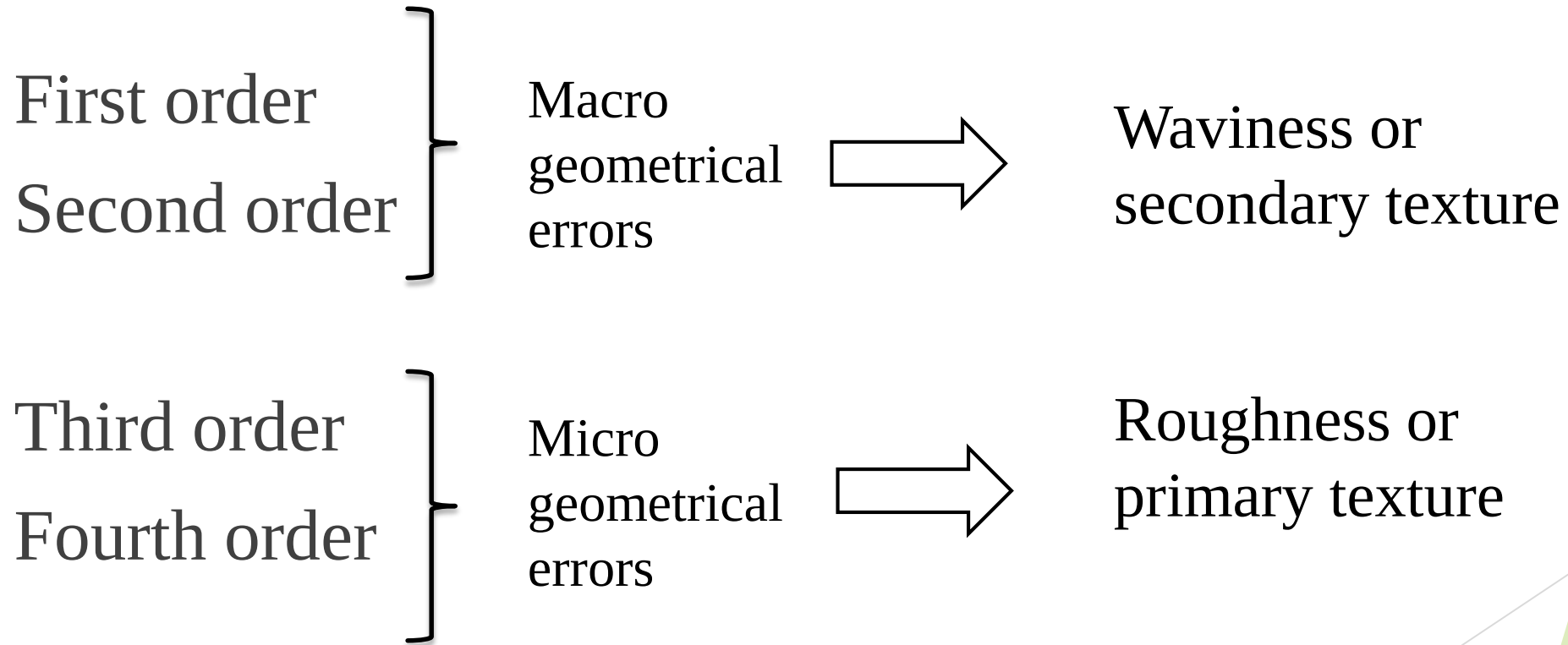
Irregularities due to vibrations of any kind

► Third order irregularity

Irregularities caused due to machining process

► Fourth order irregularity

Irregularities arising during separation of chip



Elements of surface texture

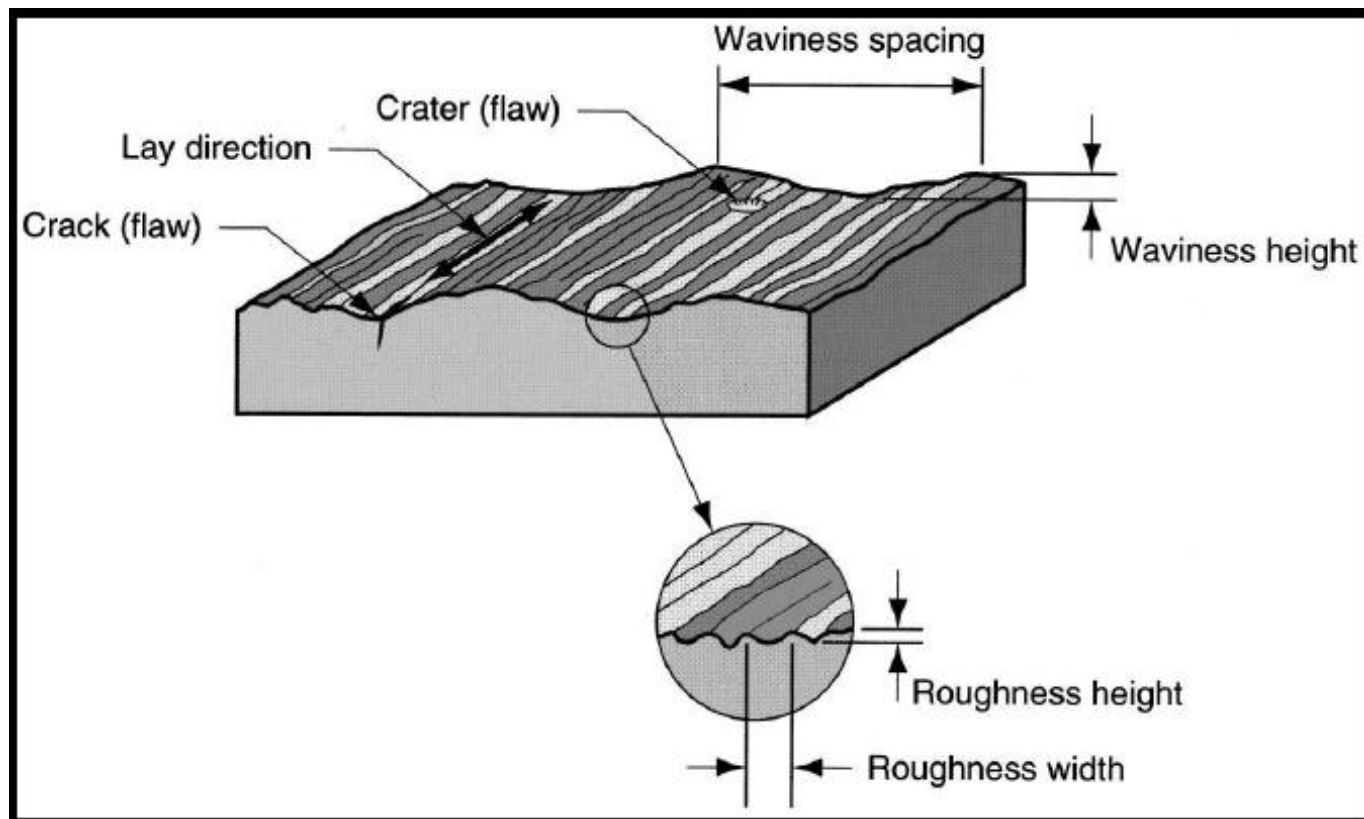
- ▶ Surface texture: repetitive or random deviation from the nominal surface. Surface texture includes

Roughness

Waviness

Lay

flaws



Roughness

- small, finely-spaced deviations from nominal surface

Waviness

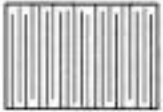
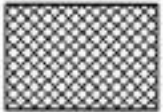
- deviations of much larger spacing

Flaws

- surface irregularities or imperfections which occur at random intervals

Lay

- it is the direction of the predominant surface pattern produced by tool marks or scratches.

Lay symbol	Surface pattern	Description
=		Lay is parallel to line representing surface to which symbol is applied.
⊥		Lay is perpendicular to line representing surface to which symbol is applied.
X		Lay is angular in both directions to line representing surface to which symbol is applied.
M		Lay is multidirectional.
C		Lay is circular relative to center of surface to which symbol is applied.
R		Lay is approximately radial relative to the center of the surface to which symbol is applied.
P		Lay is particulate, nondirectional, or protuberant.



